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SVul-Attack: Exploiting shared vulnerabilities for cross-architecture object detection attacks

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HIGHLIGHTS

- SVul-Attack exposes shared vulnerabilities in heterogeneous detectors.
- A frequency-based loss captures architecture-invariant structural features.
- A stochastic neighborhood exploration strategy for stable optimization.
- An adaptive dual-domain mechanism balancing task and feature objectives.
- SVul-Attack reduces mAP by 1.53% and LPIPS by 75.09%.

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ABSTRACT

Transfer-based adversarial attacks provide a practical way to evaluate the security of object detection under black-box conditions. However, existing methods tend to optimize surrogate-specific weaknesses, overlooking vulnerabilities shared across different detectors and thereby limiting their transferability under black-box conditions. To address this challenge, we propose SVul-Attack, a shared-vulnerability-oriented attack paradigm for object detection. SVul-Attack jointly optimizes task-level objectives and feature-level perturbations to generate architecture-invariant adversarial examples. Specifically, a frequency-aware feature loss is introduced to disrupt high-frequency structural representations in intermediate features, which are largely shared across different detection architectures. In addition, we design a stochastic neighborhood exploration strategy to approximate gradient expectations over diverse perturbation regions, thereby stabilizing the optimization process and mitigating model-specific bias. These components are integrated through an adaptive dual-domain optimization mechanism to balance attack effectiveness and generalization. Experiments on 22 object detectors show that SVul-Attack outperforms state-of-the-art methods on most evaluated detectors, achieving a further reduction of 1.53 in the average mAP, suggesting that architectural differences alone do not necessarily prevent cross-model adversarial transfer. The code will be available at https://github.com/LimingZhang719/SVul-Attack.git.

1. Introduction

With advances in visual perception technologies [1–3], object detection has become a key component of security-critical applications such as autonomous driving, intelligent surveillance, and intelligent transportation systems [4–6]. Detection failures in these

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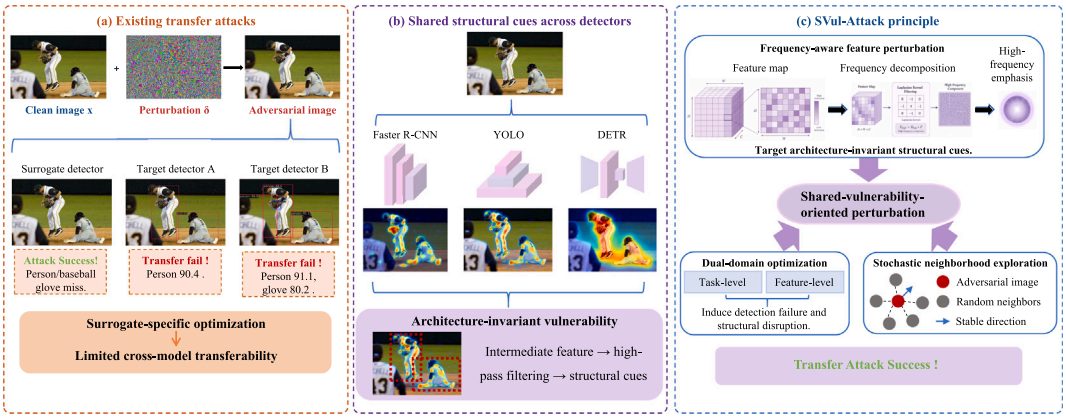


Fig. 1. Motivation and principle of SVul-Attack. (a) Existing attacks overfit surrogate-specific weaknesses and transfer poorly to heterogeneous detectors. (b) Different detectors share structural cues, revealing architecture-invariant vulnerabilities. (c) The core method of SVul-Attack.

scenarios may affect downstream security and safety decisions. Despite their importance, modern object detectors remain vulnerable to adversarial perturbations [7]. Compared with image classification, attacking object detectors is more challenging because the attacker must simultaneously disrupt localization, category prediction, and confidence estimation rather than merely induce an incorrect image-level label [8].

Transfer-based attacks offer a practical way to evaluate object detection security under black-box conditions. Existing methods improve transferability through input diversification [9,10], surrogate ensembles [11], feature manipulation [12], frequency transformation [13], or gradient stabilization [14]. However, most of them still rely on strengthening surrogate-model optimization. This may amplify surrogate-specific biases and obscure vulnerabilities shared by heterogeneous detectors, limiting cross-model generalization [15].

Rather than combining independent attack heuristics, an effective transferable attack should be designed around a unified principle: exposing vulnerabilities that are consistently shared across detector architectures while avoiding optimization paths that overfit a specific surrogate model (as shown in Fig. 1). To address this issue, we propose SVul-Attack, a shared-vulnerability-oriented paradigm for transferable attacks on object detectors. Instead of merely maximizing surrogate losses [16], SVul-Attack operationalizes this principle through three coupled components: dual-domain optimization for task-level failure and feature-level disruption, frequency-aware feature perturbation for architecture-invariant structural cues, and stochastic neighborhood exploration for stable optimization and reduced surrogate bias. Together, these components form a unified process for exposing and exploiting shared vulnerabilities under black-box conditions.

We evaluate SVul-Attack on MS COCO [17] using 22 object detectors covering two-stage, one-stage, anchor-based, anchor-free, and Transformer-based architectures. SVul-Attack achieves the best average attack performance among the compared methods, with a further reduction of 1.53 in the average mAP and a 75.09% reduction in LPIPS. These observations motivate us to explicitly identify and exploit vulnerabilities shared across heterogeneous detector architectures.

The main contributions of this paper are summarized as follows:

- We propose SVul-Attack, a shared-vulnerability-oriented transfer attack framework for object detection. By jointly optimizing task-level detection failures and feature-level structural disruption, SVul-Attack reduces surrogate-specific overfitting and improves transferability across heterogeneous detector architectures.
- We introduce a frequency-aware feature perturbation objective that emphasizes high-frequency components of intermediate representations, enabling adversarial perturbations to capture architecture-invariant structural cues shared by different detectors.
- We develop a multi-scale stochastic neighborhood exploration strategy with adaptive loss balancing to stabilize the optimization process and harmonize task-level and feature-level objectives, further improving black-box transferability and visual imperceptibility.

2. Related work

2.1. Object detection attacks

Object detection differs from image classification in that it requires simultaneous category prediction, localization, and objectness estimation for multiple objects. This multi-task nature makes adversarial attacks on detectors more challenging, especially when attacks are expected to transfer across heterogeneous architectures [8,18]. Existing attacks have explored different objectives, including suppressing objectness scores, misleading category predictions, disturbing bounding-box regression, and generating adversarial patches for physical scenarios [19–21].

Recent studies have further explored transferability from the feature, input, and attack-region perspectives. Feature- and attention-based methods perturb intermediate representations or salient regions [12,13,22], while AugTrans [9] employs object-aware input transformations to reduce surrogate overfitting. More recently, transferable background attacks [23] have shown that perturbing non-object regions can also induce detection failures across different models. These studies suggest that the choice of attack region can substantially affect cross-model transferability. However, improving transferability requires a broader consideration of how adversarial perturbations generalize across heterogeneous detector architectures, as discussed in the following section.

2.2. Transferable attacks

Transferable adversarial attacks aim to generate adversarial examples on surrogate models that remain effective on unseen target models. This property is essential for black-box attacks, where the target model is inaccessible [24]. Existing methods mainly improve transferability through input transformations, surrogate ensembles, feature disruption, attention guidance, and gradient refinement [10,14,25,26].

Input transformation methods enlarge the search space of adversarial examples by applying resizing, cropping, rotation, noise injection, or expectation over transformations [27,28]. Ensemble-based methods reduce the dependence on a single surrogate model by jointly optimizing attacks over multiple models [14,29]. Feature- and attention-guided methods disturb intermediate representations or salient regions [12,21,25], while gradient-based methods refine update directions through momentum, variance reduction, alignment, or smoothing [10,14,26,30].

These methods can improve transferability by regularizing the input, surrogate ensembles, intermediate features, or optimization directions. However, the transferable factors are generally exploited implicitly as a by-product of optimization rather than explicitly characterized across heterogeneous detector architectures. Moreover, task-level objectives, feature representations, and optimization directions are often investigated separately, leaving their shared cross-architecture properties insufficiently explored. This motivates a more explicit investigation of architecture-shared vulnerabilities for transferable object detection attacks.

2.3. Frequency-domain attacks

Frequency-domain attacks provide a structured way to analyze and generate transferable perturbations. Prior works have shown that different frequency components can affect adversarial transferability, visual quality, and robustness [31–33]. Based on this observation, frequency-based methods exploit high-frequency perturbations, low-frequency constraints, frequency masking, frequency mixing, or spatial-frequency transformations to improve attack performance [13,31,34–36]. However, most existing strategies manipulate frequency information to improve transferability without explicitly determining whether the emphasized frequency responses correspond to vulnerabilities shared across heterogeneous detectors.

Unlike these methods, we explicitly decompose shared vulnerability into task-level consensus, structural frequency-domain consensus, and optimization-direction consensus, thereby providing a unified framework for identifying and exploiting detector-invariant vulnerabilities in transferable object detection attacks.

3. Architecture-invariant adversarial attack framework

3.1. Problem definition

Let $x \in \mathbb{R}^{3\times H\times W}$ denote a clean image with ground-truth annotations $y = \{(b_i, c_i)\}_{i=1}^{N_{\text{gt}}}$, where b_i and c_i are the bounding box and category label of the i-th object, respectively. An object detector f maps x to a set of predictions:

$$\mathcal{P} = f(x) = \{(\hat{b}_j, \hat{c}_j, \hat{s}_j)\}_{j=1}^{N_{\text{pred}}} \tag{1}$$

where $\hat{b}_j$, $\hat{c}_j$, and $\hat{s}_j$ denote the predicted box, category, and confidence score, respectively.

We consider the black-box transfer attack setting for object detection. Given a set of surrogate detectors $\mathcal{F} = \{f_1, \dots, f_K\}$, the attacker generates an adversarial example:

$$x^{\text{adv}} = x + \delta \tag{2}$$

using only the surrogate detectors, while the target detectors are inaccessible during optimization and used only for evaluation. The goal is to degrade the detection performance of unseen target detectors under the perturbation constraint:

$$\|\delta\|_p \le \epsilon, \quad x^{\text{adv}} \in [0,1]^{3\times H\times W} \tag{3}$$

The key challenge in this setting is to generate perturbations that avoid overfitting to surrogate-specific weaknesses while remaining effective across heterogeneous target detectors. To aggregate optimization signals from multiple surrogates, each surrogate detector f_k is assigned a weight α_k according to its clean-image detection performance, where $\alpha_k \ge 0$ and $\sum_{k=1}^{K} \alpha_k = 1$. The overall attack objective is formulated as

$$\max_{\delta} \sum_{k=1}^{K} \alpha_k \mathcal{L}_{\text{total}}\big(f_k(x+\delta), y\big), \quad \text{s.t.} \quad \|\delta\|_p \le \epsilon,\ x+\delta \in [0,1]^{3\times H\times W} \tag{4}$$

where $\mathcal{L}_{\text{total}}$ denotes the proposed dual-domain adversarial objective that integrates task-level and feature-level losses, and is defined such that larger values indicate stronger adversarial effects.

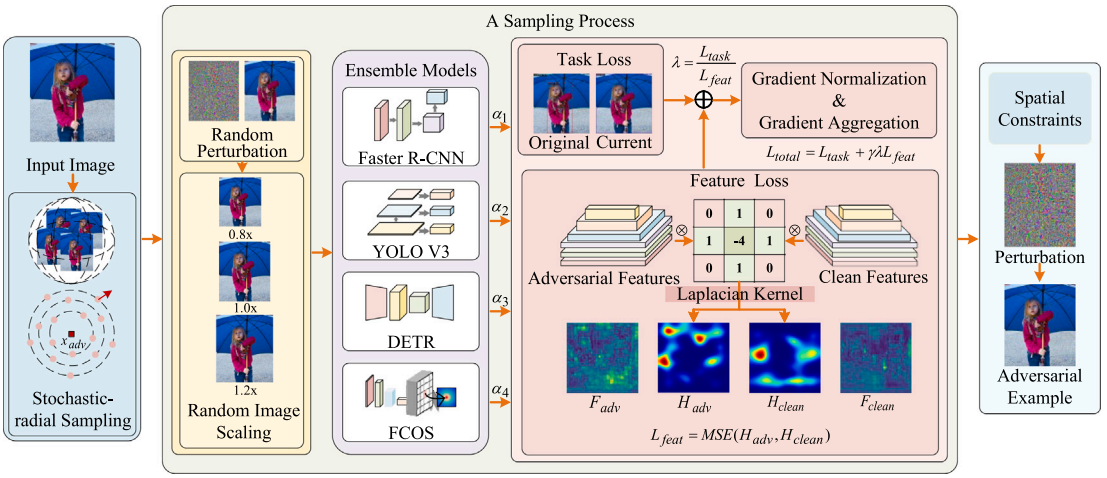


Fig. 2. Overview of the proposed dual-domain attack framework. The method jointly optimizes task-level and feature-level objectives, where α_k denotes the ensemble weight of the k-th surrogate model, λ adaptively balances their magnitudes, γ controls the feature loss intensity, F and H denote the original feature maps and the extracted high-frequency components.

3.2. Overview of shared vulnerability oriented attack

We present the proposed Shared Vulnerability-oriented Attack (SVul-Attack) framework, as illustrated in Fig. 2. The goal of SVul-Attack is to construct architecture-invariant adversarial perturbations that generalize across heterogeneous object detection models. To achieve this, we formulate the adversarial attack as a unified dual-domain optimization problem that jointly considers task-level objectives and feature-level disruption. The overall framework is built upon three key components.

First, we establish a dual-domain objective that integrates task-level and feature-level losses, enabling the attack to simultaneously degrade detection outputs and perturb intermediate representations shared across architectures (Section 3.3). In particular, the feature-level objective is instantiated in the frequency domain to capture architecture-invariant structural cues (Section 3.4). Second, to improve optimization stability and prevent overfitting to surrogate models, we introduce random image resizing and a stochastic neighborhood sampling strategy that explores diverse perturbation regions and approximates the expectation over local neighborhoods (Section 3.5). Third, we introduce a unified optimization control mechanism that combines adaptive loss balancing and spatially aware constraints to regulate both the optimization dynamics and perturbation distribution (Section 3.6). This design ensures stable convergence, balanced dual-domain optimization, and high visual imperceptibility.

Unlike conventional attacks that overfit to specific model structures, SVul-Attack explicitly targets shared representations across architectures, leading to improved cross-model transferability and visual fidelity without compromising attack efficacy.

3.3. Dual-domain formulation and task-level objective

To improve cross-model transferability, we propose a dual-domain adversarial attack framework that jointly optimizes task-level and feature-level objectives. Unlike conventional attacks that mainly attack prediction outputs, our formulation further incorporates intermediate feature perturbations to capture architecture-invariant structural representations. Formally, the overall adversarial objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \cdot \gamma \cdot \mathcal{L}_{\text{feat}} \tag{5}$$

where $\mathcal{L}_{\text{task}}$ denotes the task-level loss for degrading detection outputs, and $\mathcal{L}_{\text{feat}}$ denotes the feature-level loss for perturbing intermediate representations. γ controls the feature-level intensity and λ is an adaptive balancing coefficient defined in Section 3.6.

The task-level loss is designed to mislead the classification and localization outputs, pushing predictions away from ground-truth annotations. In contrast, the feature-level loss operates on intermediate feature maps extracted from backbone networks, encouraging consistent structural disruption across different models. By jointly optimizing these two complementary objectives, the proposed framework not only ensures strong attack effectiveness but also enhances cross-architecture transferability by targeting shared representations rather than model-specific outputs.

Given a surrogate detector set $\mathcal{F} = \{f_1, f_2, \ldots, f_K\}$, each detector f_k predicts classification scores and bounding-box locations for a set of candidate samples, such as anchors, proposals, feature points, or object queries. Let $\mathcal{A}_k$ denote the set of samples assigned by the original matching strategy of detector f_k, and let $\mathcal{A}_k^+$ denote the foreground samples matched to ground-truth objects. The task-level loss for the k-th surrogate detector is defined as:

$$\mathcal{L}_{\text{task}}^{(k)} = \frac{1}{|\mathcal{A}_k|} \sum_{i \in \mathcal{A}_k} \mathcal{L}_{\text{cls}} \left(p_{k,i}^{\text{adv}}, y_{k,i} \right) + \eta \frac{1}{|\mathcal{A}_k^+|} \sum_{i \in \mathcal{A}_k^+} \mathcal{L}_{\text{reg}} \left(b_{k,i}^{\text{adv}}, b_{k,i}^{gt} \right), \tag{6}$$

where $p_{k,i}^{\text{adv}}$ and $b_{k,i}^{\text{adv}}$ are the predicted class probability and bounding box of sample i on the adversarial image, respectively. $y_{k,i}$ and $b_{k,i}^{gt}$ are the assigned class label and matched ground-truth box obtained from the original training target assignment of detector f_k. The coefficient η balances the classification and localization terms. For different detector families, $\mathcal{A}_k$ is instantiated according to their native assignment mechanisms, including anchor matching, proposal assignment, point assignment, or query matching.

3.4. Frequency-aware feature-level objective

To instantiate the feature-level objective in the proposed framework, we design a frequency-aware feature loss. High-frequency components emphasize local structural information, such as object boundaries and textures, which play an important role in object representation. We therefore perturb these structural responses in the intermediate feature space to induce consistent representation changes across heterogeneous detectors and improve cross-model transferability.

Let $\Phi_k(\cdot)$ denote the feature extractor of the k-th surrogate detector. The multi-scale intermediate representations are defined as:

$$\mathcal{Z}_k = \Phi_k(x) = \{z_{k,l}\}_{l=1}^{L_k} \tag{7}$$

where $z_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ denotes the feature map at the l-th layer.

To extract high-frequency components, we adopt a discrete Laplacian operator, which provides a computationally efficient and spatially localized approximation. This choice aligns well with the inductive biases of convolutional architectures and enables seamless integration into iterative optimization. Specifically, we define a high-frequency extraction operator $T_{high}(\cdot)$, implemented via a depthwise convolution with a standard 3×3 Laplacian kernel K_{Lap}. The high-frequency component of feature map z_l is computed as:

$$\hat{z}_l = T_{high}(z_l) = z_l * K_{Lap} \tag{8}$$

Based on this, we define the frequency-aware feature loss as:

$$\mathcal{L}_{\text{feat}}^{(k)}(x_{\text{adv}}, x) = \sum_{l=1}^{L_k} \left\| T_{\text{high}}(\Phi_{k,l}(x_{\text{adv}})) - T_{\text{high}}(\Phi_{k,l}(x)) \right\|_2^2 \tag{9}$$

where $\|\cdot\|_2$ denotes the ℓ_2-norm and $\Phi_{k,l}(\cdot)$ represents the feature output of the l-th layer.

By maximizing the discrepancy between adversarial and clean feature representations, the proposed loss effectively disrupts structural consistency in the feature space, leading to more transferable adversarial perturbations.

3.5. Multi-scale stochastic neighborhood sampling for gradient stabilization

To effectively optimize the structural feature dual-domain objective in SVul-Attack, we introduce a Multi-scale Stochastic Neighborhood Sampling (MSNS) strategy to obtain a stable and transferable gradient estimation.

Directly optimizing the dual-domain objective based on single-point gradient evaluation is highly sensitive to local curvature and model-specific characteristics, which often leads to unstable optimization trajectories and overfitting to surrogate architectures. To address this issue, MSNS approximates the expected gradient over a local perturbation neighborhood, thereby improving robustness and transferability. Specifically, given the current adversarial example x_t, we construct N stochastic neighborhoods with increasing radii d_i. Within each neighborhood, M random perturbations $\eta_{i,j} \sim \mathcal{U}(-d_i, d_i)$ are sampled to explore diverse local regions around x_t. The gradient is then estimated by aggregating normalized gradients across all sampled points:

$$g_t = \frac{1}{M \cdot N} \sum_{i=1}^{N} \sum_{j=1}^{M} \text{Norm}(\nabla_x \mathcal{L}(f(x_t + \eta_{i,j}), y)) \tag{10}$$

where the normalization operator is defined as:

$$\text{Norm}(\nabla) = \frac{\nabla}{\|\nabla\|_p + \zeta} \tag{11}$$

where ζ is a small constant for numerical stability. This normalization eliminates magnitude discrepancies across sampled gradients, ensuring that each neighborhood contributes equally and preventing dominance by extreme gradient values.

The resulting gradient g_t can be interpreted as an approximation of the expected gradient over a local perturbation distribution, which effectively reduces estimation variance and mitigates surrogate-specific optimization bias. To further stabilize the optimization trajectory, we incorporate a momentum mechanism. The aggregated gradient is first normalized:

$$\tilde{g}_t = \frac{g_t}{\|g_t\|_1 + \zeta} \tag{12}$$

and then accumulated as:

$$m_{t+1} = \beta \cdot m_t + \tilde{g}_t \tag{13}$$

where $\beta \in [0, 1]$ denotes the momentum decay factor. Finally, the adversarial example is updated as:

$$x_{t+1} = \text{Proj}_{x,\epsilon}(x_t + \tau \cdot \text{sign}(m_{t+1})) \tag{14}$$

where τ is the step size, and $\text{Proj}_{x,\epsilon}(\cdot)$ projects the updated sample back into the ϵ-ball centered at the original input, ensuring imperceptibility constraints.

By exploring multi-scale stochastic neighborhoods and aggregating normalized gradients, MSNS provides a more reliable optimization direction for the SVul-Attack framework. Additionally, we incorporate a random image resizing strategy to further augment input diversity. Together, these approaches reduce sensitivity to local optima, prevent overfitting to surrogate models, and significantly enhance cross-model transferability.

3.6. Dynamic loss balancing and spatial constraint

To effectively optimize the dual-domain objective in SVul-Attack, it is crucial to balance the contributions of the task-level loss $\mathcal{L}_{task}$ and the feature-level loss $\mathcal{L}_{feat}$, which often exhibit significantly different numerical scales during optimization. Directly combining these losses may lead to domination by one objective, resulting in unstable optimization and suboptimal transferability. To address this issue, we introduce an adaptive loss balancing mechanism. Specifically, we define a dynamic weighting coefficient λ to automatically align the magnitudes of the two losses:

$$\mathcal{L}_{total} = \mathcal{L}_{task} + \lambda \cdot \gamma \cdot \mathcal{L}_{feat} \tag{15}$$

$$\lambda = \text{detach}\left(\frac{|\mathcal{L}_{task}|}{|\mathcal{L}_{feat}| + \zeta}\right) \tag{16}$$

where ζ is a small constant for numerical stability, and $\text{detach}(\cdot)$ indicates that λ is treated as a constant during backpropagation. This prevents the balancing factor from interfering with gradient computation, thereby stabilizing the optimization process.

The hyperparameter γ is introduced to explicitly control the relative importance of the feature-level objective. Based on the parameter sensitivity analysis described in Section 4.3.2, we set $\gamma = 2$. This observation suggests that moderately emphasizing feature-level perturbations helps capture architecture-invariant structural discrepancies without weakening the effectiveness of task-level misguidance.

In addition to loss-level balancing, we further introduce a spatial constraint mechanism to preserve visual naturalness while maintaining attack effectiveness. Instead of applying uniform perturbation constraints across all regions, we adaptively regulate perturbation intensity based on local image structures. Specifically, we define lower and upper bounds for each pixel as:

$$B_{low} = \max(P_{\min}(x, \Omega), x - \epsilon) \tag{17}$$

$$B_{high} = \min(P_{\max}(x, \Omega), x + \epsilon) \tag{18}$$

where Ω denotes the local neighborhood. The $P_{\min}$ and $P_{\max}$ operators capture local intensity variations, enabling the constraint to adapt to spatial context.

During optimization, the adversarial example is projected into the range $[B_{low}, B_{high}]$, allowing larger perturbations in texture-rich regions while restricting modifications in smooth areas. This design preserves natural visual semantics and improves perceptual quality without sacrificing attack performance.

Overall, the combination of dynamic loss balancing and spatially adaptive constraints ensures stable optimization, effective dual-domain synergy, and high visual imperceptibility in the SVul-Attack framework. The overall optimization procedure of the proposed SVul-Attack framework is summarized in Algorithm 1.

4. Experiments

4.1. Experimental settings

Datasets and Models. We conduct all experiments on the MS COCO 2017 [17] dataset, which is a widely adopted benchmark for object detection due to its large scale, diverse object categories, and complex real-world scenes. Following common practice in adversarial attack studies, we randomly sample 1000 images from the validation set to balance computational efficiency and evaluation reliability. All object detection models are implemented using the MMDetection [37] framework to ensure consistency and reproducibility. To generate transferable adversarial examples, we construct a surrogate ensemble consisting of four representative detectors spanning heterogeneous architectural paradigms: Faster R-CNN [38] (two-stage CNN-based), YOLOv3 [39] (anchor-based one-stage), FCOS [40] (anchor-free one-stage), and DETR [41] (Transformer-based). For black-box evaluation, we further include several unseen detectors, including Deformable DETR [42], RepPoints [43], and CenterNet [44], to comprehensively assess cross-model transferability. This experimental setup focuses on evaluating the ability of the proposed method to generalize across diverse model architectures rather than across datasets.

Evaluation Metrics. We evaluate attack effectiveness using standard object detection metrics, including mAP@0.5 and mAP@0.5:0.95. All results are reported in percentage form for clarity. In addition, to assess the visual quality and imperceptibility of adversarial examples, we report Structural Similarity Index (SSIM) and Learned Perceptual Image Patch Similarity (LPIPS).

All experiments were conducted on a workstation equipped with two NVIDIA GeForce RTX 4090 GPUs, using PyTorch 2.7.1 and CUDA 13.0.

Implementation Details. We constrain the adversarial perturbation under an ℓ_∞-norm bound with $\epsilon = 8/255$, following standard settings in prior work. All adversarial examples are generated with $T = 20$ iterations. For the proposed MSNS strategy, we set $N = M = 3$, and the feature scaling factor is set to $\gamma = 2$. These parameters are determined based on the parameter sensitivity analysis presented in Section 4.3.2.

Algorithm 1: SVul-Attack: Shared Vulnerability-oriented Attack.

Input: Clean image x; ground-truth labels y; surrogate detector set $\mathcal{F}$; maximum perturbation budget ϵ; iteration number T; step size τ; momentum decay factor β; multi-scale neighborhood set $\mathcal{S}$; feature loss weight γ.

Output: Adversarial example x_{adv}

Initialize $x_{\text{adv}} \leftarrow x$, accumulated gradient $m \leftarrow 0$;
for $t = 1$ **to** T **do**
  Initialize aggregated gradient $G \leftarrow 0$;
  foreach *scale* $s \in \mathcal{S}$ **do**
    Sample M perturbations $\eta_i \sim \mathcal{U}(-s, s)$, and generate $x_i = x_{adv} + \eta_i$;
    foreach *surrogate model* $f_k \in \mathcal{F}$ **do**
      Compute task loss $\mathcal{L}_{task}^{(k)}$ and feature loss $\mathcal{L}_{feat}^{(k)}$ on x_i;
      Compute adaptive weight λ;
      Construct joint loss: $\mathcal{L}^{(k)} = \mathcal{L}_{task}^{(k)} + \lambda \cdot \gamma \cdot \mathcal{L}_{feat}^{(k)}$;
      Compute normalized gradient: $g_k = \text{Norm}(\nabla_x \mathcal{L}^{(k)})$;
      Accumulate gradient: $G \leftarrow G + w_k \cdot g_k$;
    end
  end
  Normalize aggregated gradient: $G \leftarrow \frac{G}{\|G\|_1 + \xi}$;
  Update momentum: $m \leftarrow \beta \cdot m + G$;
  Update adversarial example: $x_{adv} \leftarrow x_{adv} + \alpha \cdot \text{sign}(m)$;
  Project into ϵ-ball: $x_{adv} \leftarrow \text{Proj}_{\epsilon}(x_{adv})$;
  Apply spatial constraint: $x_{adv} \leftarrow \text{clip}(x_{adv}, B_{low}, B_{high})$;
end
return x_{adv}

Table 1
Detection performance degradation under adversarial attack. Comparison of mAP@0.5 and mAP@0.5:0.95 between clean images (C) and adversarial examples (A) across different object detectors.

Description	Model	mAP @0.5-C	mAP @0.5-A	mAP @0.5:0.95-C	mAP @0.5:0.95-A
Proxy Model	Faster R-CNN	51.72	3.22	36.18	2.20
	YOLOv3	41.74	8.35	28.11	3.62
	FCOS	28.44	2.72	23.88	1.70
	DETR	58.60	3.68	41.35	2.13
Two-Stage	Libra R-CNN	53.49	4.15	37.29	2.50
	Guided Anchoring	49.29	5.17	36.53	3.23
	Grid R-CNN	52.72	4.26	39.15	2.80
	Cascade R-CNN	51.62	3.96	38.72	2.65
	GCNet	54.43	5.74	38.45	3.62
One-Stage (Anchor-Based)	RetinaNet	42.72	3.07	32.10	2.05
	FreeAnchor	54.51	5.34	39.70	3.19
	GHM	48.72	4.64	35.71	2.89
	ATSS	26.38	1.83	21.95	1.29
	YOLOF	39.31	3.52	30.29	2.33
One-Stage (Anchor-Free)	FoveaBox	41.41	3.34	30.96	2.20
	CenterNet	25.51	5.44	19.99	3.83
	RepPoints	44.15	3.75	32.60	2.54
	VFNet	56.43	7.11	43.91	4.69
Transformer-Based	Deformable DETR	44.00	3.12	34.63	2.12
	Conditional DETR	43.74	3.30	33.03	2.14
	DAB-DETR	41.61	3.23	32.16	1.99
	DINO	42.93	3.21	35.78	2.28

4.2. Comparison with state-of-the-art attacks

4.2.1. Overall attack performance

As shown in Table 1, the proposed method significantly degrades detection performance across all evaluated architectures. Specifically, the mAP values drop drastically from clean inputs to adversarial examples under both the mAP@0.5 and mAP@0.5:0.95 metrics, indicating strong attack effectiveness. More importantly, the method exhibits strong cross-model transferability. Even for unseen Transformer-based detectors such as Deformable DETR, the detection performance is reduced to extremely low levels (e.g.,

Table 2
Comparison of cross-model attack performance among different attack methods. Lower mAP@0.5 indicates stronger attack performance.

Detector	FDA	NAA	RPA	MI-FGSM	DI-FGSM	TI-FGSM	OSFD	Ours
Faster R-CNN	46.68	48.12	46.04	6.75	4.74	6.17	4.97	**3.22**
YOLOv3	38.32	39.20	39.70	42.19	13.32	13.59	21.88	**8.35**
FCOS	26.44	27.78	25.22	14.81	11.99	11.05	4.40	**2.72**
DETR	52.74	55.38	48.49	12.44	9.08	10.77	5.70	**3.68**
Libra R-CNN	47.57	48.53	49.00	4.37	8.87	8.15	5.02	**4.15**
Guided Anchoring	43.84	46.33	45.87	16.04	15.01	14.50	6.99	**5.17**
Grid R-CNN	46.68	48.90	48.78	5.33	9.27	11.28	5.66	**4.26**
Cascade R-CNN	45.94	48.23	45.47	4.07	8.07	9.24	5.48	**3.96**
GCNet	48.86	50.60	51.39	6.74	12.01	11.76	6.07	**5.74**
RetinaNet	39.61	41.07	36.64	3.88	9.25	7.57	5.03	**3.07**
FreeAnchor	49.29	50.80	49.41	6.39	10.82	11.57	5.83	**5.34**
GHM	44.39	46.49	43.53	5.84	10.45	10.12	5.22	**4.64**
ATSS	24.74	25.99	20.84	3.43	6.79	2.66	2.96	**1.83**
YOLOF	37.08	37.99	36.74	17.59	12.11	10.59	5.16	**3.52**
FoveaBox	37.50	39.26	35.27	4.57	9.35	7.88	4.68	**3.34**
CenterNet	16.34	18.71	18.82	23.90	18.47	15.96	**3.18**	5.44
RepPoints	40.13	42.03	36.45	5.93	9.87	6.33	4.24	**3.75**
VFNet	50.52	52.59	48.55	25.16	21.20	20.15	**6.49**	7.11
Deformable DETR	39.08	42.52	33.65	7.22	11.00	7.32	4.06	**3.12**
Conditional DETR	41.17	43.01	35.40	8.83	10.18	10.89	4.50	**3.30**
DAB-DETR	39.92	40.82	32.08	7.61	10.74	8.69	3.86	**3.23**
DINO	38.98	41.06	31.61	8.80	11.49	7.76	4.48	**3.21**
Avg. (All 22)	40.72	42.52	39.04	11.00	11.09	10.18	5.72	**4.19**
Avg. (18 Unseen)	40.65	42.50	38.86	9.21	11.39	10.13	4.94	**4.12**

mAP@0.5 = 3.12%), demonstrating that the generated perturbations generalize well across heterogeneous architectures. This behavior suggests that the proposed dual-domain design effectively disrupts shared structural representations rather than overfitting to specific detection heads. At the same time, the generated adversarial examples maintain high visual quality, achieving an average LPIPS of 0.1010 and an SSIM of 0.906. This indicates that the attack is not only effective but also visually imperceptible.

4.2.2. Comparison with state-of-the-art attacks

We further compare the proposed method with representative transferable attacks, including MI-FGSM [45], DI-FGSM [46], TI-FGSM [47], FDA [48], NAA [49], RPA [50], and OSFD [16]. For a fair comparison, all methods are evaluated using the same surrogate ensemble, perturbation budget, iteration number, dataset, and evaluation protocol. As summarized in Tables 2 and 3, our method achieves the strongest overall attack performance while maintaining competitive image quality.

Analysis of CenterNet and VFNet. Although SVul-Attack achieves the best overall performance, OSFD obtains lower mAP on CenterNet and VFNet. A plausible explanation is that their specialized prediction mechanisms, including keypoint-based feature aggregation in CenterNet and IoU-aware classification with feature refinement in VFNet, exhibit different sensitivities to feature perturbations, while OSFD's object- and boundary-oriented feature disruption may better match these representations. Nevertheless, our ablation results show that $\mathcal{L}_{\text{feat}}$ and MSNS remain effective on both detectors, indicating that these exceptions are mainly associated with model-specific response characteristics.

Analysis of YOLOv3. Although YOLOv3 is included in the surrogate ensemble, its performance degradation is comparable to that of certain black-box models. We attribute this to its dense multi-scale prediction mechanism, which reduces sensitivity to localized perturbations. This observation further supports the conclusion that the proposed method prioritizes generalizable structural disruption rather than overfitting to specific models.

4.2.3. Visual quality analysis

Beyond attack effectiveness, visual imperceptibility is another important criterion for evaluating adversarial attacks. We therefore separately analyze the visual quality of adversarial examples generated by different methods. This section evaluates whether SVul-Attack improves cross-model transferability without introducing perceptible artifacts, using both quantitative metrics and qualitative visual comparisons.

Quantitative comparison. As summarized in Table 3, compared with feature-level attack methods such as OSFD, which often introduce noticeable visual distortions (LPIPS > 0.4), our method produces significantly more natural adversarial examples. The lower LPIPS (0.1010) and higher SSIM (0.906) indicate a better balance between attack strength and visual imperceptibility.

Qualitative visualization. As illustrated in Fig. 3, when attacking challenging models such as Deformable DETR, existing methods often produce visible artifacts, whereas our method generates visually natural images while effectively suppressing detection results. This demonstrates that the proposed framework achieves a favorable trade-off between transferability and imperceptibility.

Table 3
Comparison of visual imperceptibility among different attack methods. Lower LPIPS and higher SSIM indicate better visual quality.

Metric	FDA	NAA	RPA	MI-FGSM	DI-FGSM	TI-FGSM	OSFD	Ours
LPIPS ↓	0.3318	0.2704	0.4016	0.1231	0.1154	0.1300	0.4055	**0.1010**
SSIM ↑	0.4558	0.4920	0.4382	0.8718	0.8696	0.8614	0.4361	**0.9060**



Fig. 3. Visual comparison of adversarial examples on Deformable DETR. Our method generates visually imperceptible perturbations while effectively suppressing object detections, compared to baseline attacks that introduce noticeable artifacts.

4.3. Ablation and parameter sensitivity

4.3.1. Ablation study

To evaluate the contribution of each component in the proposed SVul-Attack framework, we conduct ablation studies by systematically removing four key modules: the feature-level objective (L_{feat}), the task-level objective (L_{task}), the multi-scale stochastic neighborhood sampling strategy (MSNS), and the spatial constraint (SC). The corresponding variants are denoted as w/o L_{feat}, w/o L_{task}, w/o MSNS, and w/o SC, respectively. The full SVul-Attack model is denoted as Full. The quantitative results are summarized in Tables 4 and 5.

Effect of Feature-level Objective. As shown in Table 4, removing the feature-level loss (w/o $\mathcal{L}_{\text{feat}}$) leads to a consistent and significant increase in mAP across all evaluated models, indicating a substantial drop in attack effectiveness. For example, the mAP on CenterNet increases from 5.44% to 13.49%, and on RetinaNet from 3.07% to 7.06%, with similar trends observed across Transformer-based detectors such as DETR and DINO. This confirms that the feature-level objective is a key driver of transferability, as it explicitly disrupts shared intermediate representations that are largely architecture-invariant.

Effect of Task-Level Objective. The task-level objective plays a direct role in degrading the detection outputs by misleading both classification and localization predictions. As shown in Table 4, removing this component (w/o $\mathcal{L}_{\text{task}}$) leads to noticeable performance degradation across several models. For example, the mAP increases from 3.96% to 5.53% on Cascade R-CNN and from 3.52% to 4.79% on YOLOF. This indicates that relying solely on feature-level perturbation is insufficient to consistently suppress the detection results. The task-level objective provides explicit supervision for attacking the prediction outputs, thereby complementing the feature-level objective and improving overall attack effectiveness.

Effect of Multi-scale Stochastic Neighborhood Sampling. Removing multi-scale stochastic neighborhood exploration (w/o MSNS) results in a noticeable degradation in attack performance across most models. For instance, the mAP on FCOS increases from

Table 4
Ablation study of attack performance on different object detectors.

Detector	w/o $\mathcal{L}_{\text{feat}}$	w/o $\mathcal{L}_{\text{task}}$	w/o MSNS	w/o SC	Full
Faster R-CNN	4.84	3.91	**3.15**	2.53	3.22
YOLOv3	13.32	**8.30**	9.25	5.67	8.35
FCOS	6.42	3.28	8.21	1.93	**2.72**
DETR	7.76	4.96	5.71	2.96	**3.68**
Libra R-CNN	7.81	4.96	5.12	2.51	**4.15**
Guided Anchoring	10.75	6.71	11.74	3.95	**5.17**
Grid R-CNN	8.10	5.12	6.54	2.91	**4.26**
Cascade R-CNN	7.70	5.53	5.67	2.81	**3.96**
GCNet	10.02	6.47	8.56	3.50	**5.74**
RetinaNet	7.06	3.73	4.64	2.50	**3.07**
FreeAnchor	9.63	6.29	8.09	3.71	**5.34**
GHM	9.21	5.48	6.54	3.31	**4.64**
ATSS	4.75	2.12	1.85	1.02	**1.83**
YOLOF	8.16	4.79	7.90	2.66	**3.52**
FoveaBox	6.99	4.34	4.72	2.50	**3.34**
CenterNet	13.49	6.36	16.20	3.68	**5.44**
RepPoints	7.42	4.16	3.78	2.14	**3.75**
VFNet	15.97	8.51	17.59	6.03	**7.11**
Deformable DETR	7.85	4.35	5.37	2.82	**3.12**
Conditional DETR	7.99	3.88	6.45	2.85	**3.30**
DAB-DETR	7.63	3.93	5.96	2.67	**3.23**
DINO	7.67	3.94	5.12	2.74	**3.21**
Avg. (All 22)	8.66	5.05	7.19	3.06	**4.19**
Avg. (18 Unseen)	8.79	5.04	7.32	3.02	**4.12**

Table 5
Ablation study of visual quality. Lower LPIPS and higher SSIM indicate better visual imperceptibility.

Metric	w/o L_{feat}	w/o $\mathcal{L}_{\text{task}}$	w/o MSNS	w/o SC	Full
LPIPS ↓	0.1684	0.1021	0.1629	0.1988	0.1010
SSIM ↑	0.9025	0.9053	0.9032	0.8183	0.9060

2.72% to 8.21%, and on VFNet from 7.11% to 17.59%. This demonstrates that aggregating gradients over multiple local neighborhoods effectively reduces optimization variance and prevents overfitting to specific surrogate models, thereby improving generalization.

Effect of Spatial Constraints. The spatial constraint is not intended to maximize attack strength alone; instead, it regularizes the perturbation distribution to improve perceptual quality. Therefore, the full model is evaluated as a trade-off between transferability and imperceptibility. Removing spatial constraints (w/o SC) slightly improves attack effectiveness, as reflected by lower mAP values on several models (e.g., from 3.07% to 2.50% on RetinaNet). However, this comes at the cost of significantly degraded visual quality (as shown in Table 5).

To further validate this observation, Fig. 4 visualizes the intermediate feature responses and attention distributions under different settings. Compared to task-only optimization, the proposed dual-domain method produces more dispersed feature activations and significantly attenuated attention regions. This indicates that the model's internal structural representations are effectively corrupted, which explains the improved cross-model transferability observed in Table 4.

4.3.2. Parameter sensitivity

MSNS Sampling Parameters. For the proposed MSNS, we determine the sampling configuration via grid search. Specifically, for each candidate combination of number of scales N and sampling frequency M, we perform leave-one-surrogate-out validation over the four surrogate detectors, where three detectors are used for attack optimization and the remaining detector is held out for transfer evaluation. This process is repeated for all four detectors, and the configuration is evaluated based on the sum of the four held-out mAP values, as shown in Fig. 5. The results show that $N = 3$ and $M = 3$ achieve the lowest cumulative held-out mAP among the evaluated configurations. Meanwhile, increasing N and M leads to higher computational costs due to the increased number of gradient evaluations, while stronger sampling does not yield monotonic improvements in transferability. Therefore, we adopt $N = 3$ and $M = 3$ as the default configuration, which provides a favorable trade-off between attack effectiveness and computational efficiency.

Feature Scaling Factor. The feature scaling factor γ, which controls the relative importance of the feature-level objective in SVul-Attack, is tuned using the Optuna hyperparameter optimization framework and is set to $\gamma = 2$ in all experiments. This setting is found to provide the best balance between attack strength and cross-model transferability.

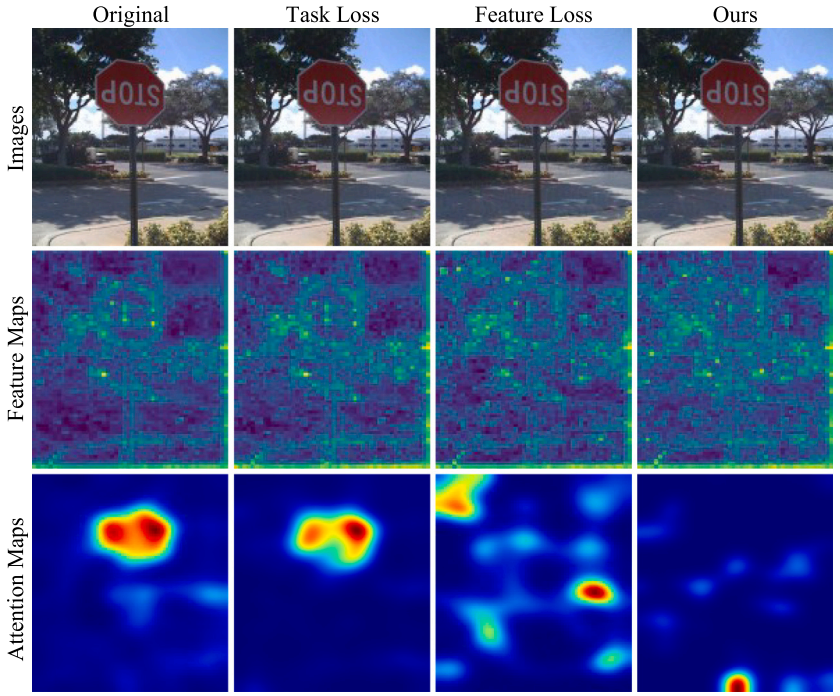


Fig. 4. Visualization of intermediate feature maps and attention distributions under different attack settings.

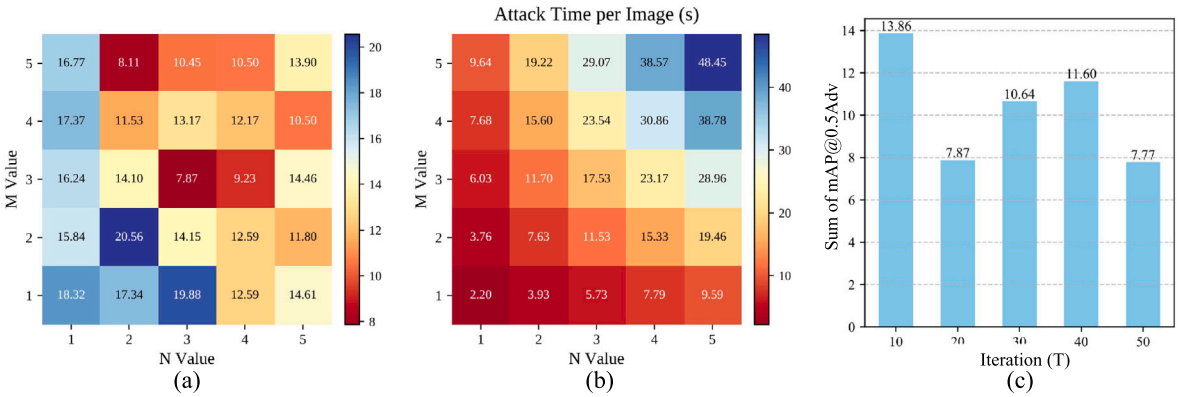


Fig. 5. Sensitivity and efficiency analysis of MSNS sampling and iteration parameters.

Iteration Number. We further analyze the impact of the iteration budget T. As illustrated in Fig. 5, T = 20 already achieves near-optimal attack performance, while further increasing the iteration budget does not yield consistent improvements and incurs additional computational cost.

Perturbation Budgets. We evaluate different perturbation budgets of $\epsilon = 4/255, 8/255,$ and $16/255$ on all 22 evaluated detectors. As shown in Fig. 6, our method consistently achieves the lowest average mAP across all budgets. At $\epsilon = 4/255$, it achieves an average mAP of 6.34, 2.24 points lower than OSFD, and the advantage remains at larger budgets, reaching 0.27 mAP at $\epsilon = 16/255$. These results demonstrate stable effectiveness across different perturbation budgets.

Random Seeds. We repeat the experiments using three random seeds (0, 1 and 2) under the default attack setting. Across 22 evaluated detectors, our method achieves an average mAP of 4.18 ± 0.03, outperforming OSFD (5.80 ± 0.04) with only minor variation across runs, demonstrating stable performance under different random seeds.

4.4. Shared-vulnerability mechanism analysis

To investigate shared vulnerable representations across heterogeneous detectors, we conduct centered kernel alignment (CKA) analysis on the intermediate representations of Libra R-CNN [51], YOLOF [52], RepPoints [43], and DINO [53]. As shown in Table 6, the average CKA values for clean inputs are 0.20, 0.25, and 0.33 for the low-, middle-, and high-frequency representations, respectively. Under adversarial inputs, these values increase to 0.35, 0.44, and 0.70, corresponding to alignment gains of 0.14, 0.19, and

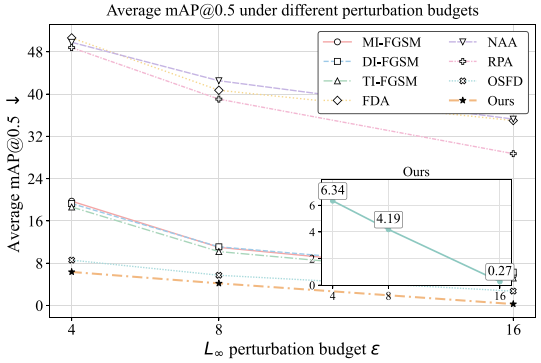


Fig. 6. Attack performance under different perturbation budgets.

Table 6
Cross-architecture CKA similarity of low-, middle-, and high-frequency representations under clean and adversarial inputs. Δ denotes the increase in CKA from clean to adversarial representations.

Model Pair	Clean			Adversarial			Δ		
	Low	Middle	High	Low	Middle	High	Low	Middle	High
Libra R-CNN – YOLOF	0.21	0.34	0.50	0.31	0.58	0.81	0.10	0.24	0.31
Libra R-CNN – RepPoints	0.20	0.36	0.45	0.36	0.51	0.75	0.15	0.15	0.31
Libra R-CNN – DINO	0.32	0.34	0.46	0.61	0.66	0.84	0.30	0.32	0.39
YOLOF – RepPoints	0.14	0.16	0.15	0.20	0.31	0.50	0.06	0.15	0.35
YOLOF – DINO	0.10	0.08	0.09	0.20	0.19	0.54	0.10	0.11	0.45
RepPoints – DINO	0.25	0.20	0.31	0.41	0.38	0.74	0.16	0.18	0.42
Average	0.20	0.25	**0.33**	0.35	0.44	**0.70**	0.14	0.19	**0.37**

Table 7
Comparison of attack performance using different feature representations on different object detectors.

Detector	Raw	Low	Middle	High
Libra R-CNN	4.90	5.93	6.26	**4.15**
Guided Anchoring	7.81	7.97	7.62	**5.17**
Grid R-CNN	5.43	5.61	6.24	**4.26**
Cascade R-CNN	5.75	5.69	6.07	**3.96**
GCNet	6.96	7.39	8.19	**5.74**
RetinaNet	3.60	5.33	4.55	**3.07**
FreeAnchor	6.88	7.15	6.96	**5.34**
GHM	5.52	5.94	6.04	**4.64**
ATSS	3.11	3.86	3.86	**1.83**
YOLOF	4.88	5.99	5.85	**3.52**
FoveaBox	4.69	5.48	6.17	**3.34**
CenterNet	6.55	8.71	7.85	**5.44**
RepPoints	3.96	4.78	5.52	**3.75**
VFNet	9.95	10.09	10.26	**7.11**
Deformable DETR	3.78	5.25	4.35	**3.12**
Conditional DETR	4.70	5.21	5.21	**3.30**
DAB-DETR	4.76	5.46	4.73	**3.23**
DINO	3.60	4.35	4.23	**3.21**
Average	**5.38**	**6.12**	**6.11**	**4.12**

0.37. The substantially increased cross-architecture alignment under adversarial perturbations provides representation-level evidence of shared vulnerable responses across heterogeneous detectors. Moreover, the strongest alignment and largest increase are observed in the high-frequency representations, indicating that such responses are more consistently affected across different architectures.

Effect of Frequency Components on Transferability. We further evaluate the relationship between frequency components and attack transferability by separately optimizing low-, middle-, and high-frequency representations while keeping all other settings unchanged. As shown in Table 7, the high-frequency variant achieves the lowest average mAP@0.5 of 4.12 across the 18 unseen

Table 8
Effect of the number and composition of surrogate models on attack transferability.

Surrogates		Avg. mAP@0.5 ↓
1	Faster R-CNN	4.17
	YOLOv3	18.30
	FCOS	12.16
	DETR	13.76
	Mean	12.10
2	Faster R-CNN + YOLOv3	5.86
	Faster R-CNN + FCOS	3.87
	Faster R-CNN + DETR	4.09
	YOLOv3 + FCOS	10.71
	YOLOv3 + DETR	11.08
	FCOS + DETR	8.99
	Mean	7.43
3	Faster R-CNN + YOLOv3 + FCOS	4.81
	Faster R-CNN + YOLOv3 + DETR	4.99
	Faster R-CNN + FCOS + DETR	3.56
	YOLOv3 + FCOS + DETR	7.21
	Mean	5.14
4	Faster R-CNN + YOLOv3 + FCOS + DETR	4.12

Table 9
Effect of surrogate architectural diversity on attack transferability.

Ensemble Type	Surrogate Configuration	Avg. mAP@0.5 ↓
Homogeneous	Faster + Libra + Grid + Cascade	3.45
	YOLOv3 + RetinaNet + FreeAnchor + YOLOF	5.91
	FCOS + FoveaBox + RepPoints + VFNet	3.92
	DETR + Conditional + DAB + DINO	5.32
	Mean ± Std	4.65 ± 1.16
Heterogeneous	Faster + YOLOv3 + FCOS + DETR	4.32
	Libra + RetinaNet + FoveaBox + Conditional DETR	3.98
	Grid + FreeAnchor + RepPoints + DAB-DETR	4.32
	Cascade + YOLOF + VFNet + DINO	4.01
	Mean ± Std	4.16 ± 0.19

detectors, compared with 6.12 and 6.11 for the low- and middle-frequency variants, respectively. These results show that high-frequency structural responses not only exhibit stronger cross-architecture adversarial alignment but also provide a more transferable attack target, supporting the frequency-aware feature perturbation adopted in SVul-Attack.

4.5. Surrogate configuration analysis

To disentangle the effect of surrogate ensemble design from the contribution of the proposed method, we conduct two controlled evaluations. First, we analyze the surrogate number and composition by evaluating all subsets of the four original surrogates on the same 18 unseen detectors, with Faster R-CNN, YOLOv3, FCOS, and DETR excluded from the target set. Second, we isolate the effect of architectural diversity by comparing four homogeneous and four heterogeneous four-model ensembles on the same six held-out detectors: Guided Anchoring, GCNet, GHM, ATSS, CenterNet, and Deformable DETR. None of these detectors participates in any of the compared ensembles. Each homogeneous ensemble represents one detector family, whereas each heterogeneous ensemble contains one detector from each architectural family.

Effect of Surrogate Number and Composition. As shown in Table 8, the mean mAP decreases from 12.10 to 7.43, 5.14, and 4.12 as the number of surrogates increases from one to four, showing that more surrogates generally improve transferability. However, model composition also plays an important role. For $K = 2$, mAP ranges from 3.87 to 11.08, whereas the Faster R-CNN + FCOS + DETR ensemble achieves 3.56 mAP, outperforming the four-surrogate ensemble with 4.12. Thus, transferability is jointly affected by surrogate number and composition rather than surrogate number alone.

Effect of Surrogate Diversity. As shown in Table 9, heterogeneous ensembles achieve an average mAP of 4.16 ± 0.19, compared with 4.65 ± 1.16 for homogeneous ensembles, indicating that architectural diversity is beneficial on average. However, the best homogeneous ensemble achieves an mAP of 3.45, showing that diversity alone does not determine attack effectiveness. As reported in Table 4, with the original surrogate ensemble fixed, removing $\mathcal{L}_{\text{feat}}$ or MSNS increases the average mAP on the 18 unseen detectors from 4.12 to 8.79 and 7.32, respectively. Taken together, these results show that surrogate ensemble design affects transferability, but cannot fully account for the performance gains of SVul-Attack, which are substantially contributed by the proposed mechanisms.

Table 10
Computational cost and attack effectiveness comparison.

Method	Avg. mAP@0.5 ↓	Time (s/img) ↓	Peak Memory (GB) ↓
MI-FGSM	11.0	1.3	7.6
DI-FGSM	11.1	1.9	7.7
TI-FGSM	10.2	2.7	7.9
FDA	40.7	2.1	8.3
NAA	42.5	3.8	9.0
RPA	39.0	5.3	8.5
OSFD	5.7	3.4	9.0
SVul w/o MSNS	7.2	2.2	9.1
SVul-Attack	4.2	17.5	9.2

4.6. Computational efficiency analysis

MSNS improves gradient stability by aggregating optimization directions from multiple stochastic neighborhoods, at the cost of additional forward and backward computations. Table 10 reports the average attack time per image and peak GPU memory consumption of different attacks under the same experimental setup. We additionally include SVul-Attack without MSNS to isolate the computational overhead introduced by neighborhood sampling.

Averaged over all 22 detectors, removing MSNS results in an mAP@0.5 of 7.2%, whereas the full SVul-Attack reduces it to 4.2%, corresponding to a 3.0-point reduction in mAP. This improvement is accompanied by an increase in average attack time from 2.2 to 17.5 s/image and in peak GPU memory from 9.1 to 9.2 GB. These results explicitly quantify the additional computational cost introduced by MSNS together with its contribution to cross-model transferability.

The proposed attack is primarily intended as an offline security evaluation tool rather than a real-time inference method; therefore, the additional optimization cost is acceptable when stronger black-box attack effectiveness is prioritized.

5. Conclusion

In this paper, we propose SVul-Attack, a transferable adversarial attack framework for object detection that jointly integrates stochastic neighborhood exploration with dual-domain optimization over task-level and frequency-aware feature-level objectives. This design effectively mitigates surrogate overfitting and enables the construction of architecture-invariant perturbations. Extensive experiments demonstrate that SVul-Attack achieves strong cross-model transferability and attack effectiveness across diverse detection architectures while maintaining high visual imperceptibility. Furthermore, our results suggest that diversity in detection architectures alone may be insufficient to guarantee robustness against transferable attacks. Model robustness can also be affected by factors such as training strategies, feature representations, input processing, and dedicated adversarial defenses, which are beyond the scope of the present study.

CRediT authorship contribution statement

Liyan Shen: Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Liming Zhang:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Jun Li:** Writing – review & editing, Resources, Funding acquisition, Conceptualization. **Yawei Ren:** Writing – review & editing, Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets used in this study are publicly available from their original sources. The code is available at:https://github.com/LimingZhang719/SVul-Attack.git.

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